

CT3: CSE 201 - Object-Oriented Programming I: Java (Sec C)

Time: 30 min

Mark: 20

Name:

ID:

1. Create a sub class of the following “**Tiger**” class. Name the subclass as “**SiberianTiger**”. [9]
In the “**SiberianTiger**” class, implement the following:
 - a. Create a constructor in proper way.
 - b. Create an overridden method of parent’s **hunt()** method.
 - c. Create an overloaded method of parent’s **hunt()** method.

```
public class Tiger {  
    String type, color;  
  
    public Tiger(String type, String color) {  
        this.type = type;  
        this.color = color;  
    }  
  
    public void hunt() {  
        System.out.println("Tiger hunts.");  
    }  
}
```

2. Create a class name “**Zoo**” and add main method. Inside the main method, implement the following: [9]
 - a. Create an object of “**SiberianTiger**” and assign to a reference variable “**tiger**” of “**Tiger**” type.
 - b. Call “**hunt()**” method using the **tiger** variable.
 - Will the method execute successfully or show error? A) If successful, what will be the output. B) If show error, what will be the error and how can you fix that.
 - c. Call the overloaded “**hunt(..)**” (Q1d) method using the **tiger** variable.
 - Will the method execute successfully or show error? A) If executes successfully, what will be the output. B) If shows error, what will be the error and how can you fix that.
3. Assume class **Student** has a **private float** type variable **cgpa**. How can you convert this to a read only variable? [2]

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1. Create a sub class of the following “**Vehicle**” class. Name the subclass as “**Car**”. In the “**Car**” class, implement the following: [9]
 - a. Create a constructor in proper way. Also set the **hasEngine** to **true**.
 - b. Create an overridden method of parent’s **run()** method.
 - c. Create an overloaded method of parent’s **run()** method.

```
public class Vehicle {  
    String model;  
    boolean hasEngine;  
  
    public Vehicle(String model, boolean hasEngine) {  
        this.model = model;  
        this.hasEngine = hasEngine;  
    }  
  
    public void run() {  
        System.out.println("Running");  
    }  
}
```

2. Create a class name “**TestVehicle**” and add main method. Inside the main method, implement the following: [9]
 - a. Create an object of “**Car**” and assign to a reference variable “**car**” of “**Vehicle**” type.
 - b. Call “**run()**” method using the **car** variable.
 - Will the method execute successfully or show error? A) If successful, what will be the output. B) If show error, what will be the error and how can you fix that.
 - c. Call the overloaded “**run(..)**” (Q1d) method using the **car** variable.
 - Will the method execute successfully or show error? A) If executes successfully, what will be the output. B) If shows error, what will be the error and how can you fix that.

3. Assume **Vehicle** class has 2 subclasses: **Car** and **Bicycle**. The following method is in **Test** class, the method currently takes **Bicycle** as parameter. Update the method parameter (should have only one parameter) so that it can take both **Car** or **Bicycle**. [2]

```
public static void run(Bicycle b) {  
  
}
```

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1. Create a sub class of the following “**Vehicle**” class. Name the subclass as “**Car**”. In the “**Car**” class, implement the following: [10]
 - d. Add an attribute name “noOfDoor”
 - e. Create a constructor in proper way.
 - f. Create an overridden method of parent’s **run(int speed)** method.
 - g. Create an overloaded method of parent’s **run(int speed)** method.

```
public class Vehicle {
    String make, model;
    float enginePower;

    public Vehicle(String make, String model, float engPower) {
        this.make = make;
        this.model = model;
        this.enginePower = engPower;
    }

    public void run(int speed) {
        System.out.println("Running at speed:" + speed);
    }
}
```

2. Create a class name “**TestVehicle**” and add main method. Inside the main method, [10] implement the following:
 - d. Create an object of “**Car**” and assign to a reference variable “**car**” of “**Vehicle**” type.
 - e. Call “**run(...)**” method using the **car** variable and pass the last 2 digits of you registration number as the value of **speed** parameter.
 - Will the method execute successfully or show error? A) If successful, what will be the output. B) If show error, what will be the error and how can you fix that.
 - f. Call the overloaded “**run(..)**” (Q1d) method using the **car** variable.
 - Will the method execute successfully or show error? A) If executes successfully, what will be the output. B) If shows error, what will be the error and how can you fix that.

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1. Create a sub class of the following “**Student**” class. Name the subclass as [10]
“**GraduateStudent**”. In the “**GraduateStudent**” class, implement the following:
 - a. Add an attribute name “supervisor”
 - b. Create a constructor in proper way.
 - c. Create an overridden method of parent’s **study()** method.
 - d. Create an overloaded method of parent’s **study()** method.

```
public class Student {  
    String name, id;  
    private float cgpa;  
  
    public Student(String name, String id, float cgpa) {  
        this.name = name;  
        this.id = id;  
        this.cgpa = cgpa;  
    }  
  
    public void study() {  
        System.out.println("Studying");  
    }  
}
```

2. Create a class name “**TestStudent**” and add main method. Inside the main method, [10]
implement the following:
 - a. Create an object of “**GraduateStudent**” and assign to a reference variable “**st**” of
“**Student**” type.
 - b. Call “**study()**” method using the **st** variable.
 - Will the method execute successfully or show error? A) If successful, what will
be the output. B) If show error, what will be the error and how can you fix that.
 - c. Call the overloaded “**study(..)**” (Q1d) method using the **st** variable.
 - Will the method execute successfully or show error? A) If executes successfully,
what will be the output. B) If shows error, what will be the error and how can
you fix that.

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1. Create a sub class of the following “**Performer**” class. Name the subclass as “**Singer**”. In the “**Singer**” class, implement the following: [8]
 - e. Create a constructor in proper way. Make sure to pass “**music**” for **field**.
 - f. Create an overridden method of parent’s **perform()** method.

```
public class Performer {
    String name, field;

    public Performer(String name, String field) {
        this.name = name;
        this.field = field;
    }

    public void perform() {
        System.out.println("Performing...");
    }
}
```

2. Create a class name “**CulturalShow**” and add main method. Inside the main method, implement the following: [10]
 - d. Create an object of “**Performer**” and assign to a reference variable “**perf1**” of “**Performer**” type. Pass your name and **acting** as the value of **name** and **field**.
 - e. Create an object of “**Singer**” with proper values and assign to a reference variable “**perf2**” of “**Performer**” type.
 - f. Create an object of “**Singer**” with proper values and assign to a reference variable “**perf3**” of “**Singer**” type.
 - g. Call “**perform()**” method using the **perf1** variable.
 - a. what will be the output of this call?
 - h. Call the “**perform()**” method using the **perf2** variable.
 - a. what will be the output of this call?
 - i. Call the “**perform()**” method using the **perf3** variable.
 - a. what will be the output of this call?

3. Assume the **Voter** class has the following 2 constructors. Rewrite the body of second constructor and call/invoke the first constructor such that only the **voterId** gets initialized, maintaining the behaviour of the original version. [2]

```
// First Constructor
public Voter(String name, String voterId, int age) {
    this.name = name;
    this.voterId = voterId;
    this.age = age;
}
// Second Constructor
public Voter(String voterId) {
    this.voterId = voterId;
}
```

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1. Create a sub class of the following “**Bird**” class. Name the subclass as “**Emu**”. It is worth [9]
noting that Emu is a bird that cannot fly. Having said that, in the “**Emu**” class, implement the following:
 - a. Create a constructor in proper way. Make sure to pass **false** for **canFly**.
 - b. Create an overridden method of parent’s **fly()** method.

```
public class Bird {  
    String name;  
    boolean canFly;  
    public Bird(String name, boolean canFly) {  
        this.name = name;  
        this.canFly = canFly;  
    }  
  
    public void fly() {  
        System.out.printf("%s can fly.", name);  
    }  
}
```

2. Create a class name “**Zoo**” and add main method. Inside the main method, implement [9]
the following:
 - a. Create an object of “**Bird**” class and assign to a reference variable “**bird1**” of “**Bird**” type. Pass “**Sparrow**” and **true** as the value of **name** and **canFly**.
 - b. Create an object of “**Emu**” and assign to a reference variable “**bird2**” of “**Emu**” type.
 - c. Create an object of “**Emu**” and assign to a reference variable “**bird3**” of “**Bird**” type.
 - d. Call “**fly()**” method using the **bird1** variable.
 - what will be the output of this call?
 - e. Call the “**fly()**” method using the **bird2** variable.
 - what will be the output of this call?
 - f. Call the “**fly()**” method using the **bird3** variable.
 - what will be the output of this call?
3. **Declare** 2 overloaded methods for the method below. **Note:** You do not need to [2]
implement the body of the methods.

```
public void submitAssignment(String courseCode)
```